

Thermographs and Newton polygons are two tropical objects

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Abstract

This note resolves `docs/OPEN.md` problem `under*(e_g^e_s)`: whether the two tropical consumers in `ogdoad`—game thermography and valuation/Newton polygons—are one tropical object or merely a convention mirror. They are two objects, connected by a genuine valuation shadow and external filtered transports, but there is no faithful Newton-style dyadic graded ring on the game side.

The negative theorem is small but decisive: the thermograph is not a congruence for disjunctive sum. There is no binary operation, single- or multi-step, that takes only $\text{Th}(G)$ and $\text{Th}(H)$ and returns $\text{Th}(G + H)$ for all short games. The witness is already in the temperature-zero layer: $\text{Th}(\ast) = \text{Th}(\uparrow)$, but $\ast + \ast = 0$ while $\uparrow + \ast$ is still a non-number of temperature 0.

The useful replacement is not “no tropical structure”. Temperature itself has exactly the expected tropical-valuation shape: $\text{temp}(G + H) \leq \max(\text{temp } G, \text{temp } H)$, with equal-temperature pairs forming the game-side vanishing locus. What fails is the proposed angular component $(\text{mean}(G), \text{temp}(G))$: it forgets the leading thermic residue. The honest replacement begins with the associated graded object of the game group by temperature, whose leading class retains precisely that missing residue.

The second pass, after `src/games/heating.rs` shipped game-valued heating, Berlekamp overheating, and Norton multiplication, tests the multiplicative hope directly. The unrestricted answer is negative. In the $\tau = 0$ quotient, $\ast$ and $\ast + 1$ differ by a cold number, but Norton multiplication by the positive infinitesimal unit $\uparrow$ converts that hidden integer into a leading temperature-zero residue. The same obstruction appears for the degenerate overheating operator $\int_{\uparrow}^0$. Thus the full Berlekamp/Norton operator does not descend to the naive associated graded.

The third pass settles the leading restricted case positively. If the Norton unit is the positive dyadic number $u = m/2^k$, put $\delta = 2^{-k}$ (and $\delta = 1$ for an integer) and $a = u - \delta$. Then every nonnumeric thermographic game obeys

$$\text{mean}(G.u) = u \text{mean}(G), \quad \text{temp}(G.u) = u \text{temp}(G) + a.$$

A cold dyadic number can become hot under Norton multiplication, but its exact temperature is always strictly below a . Norton linearity therefore induces an additive map $\text{gr}_{\tau} \rightarrow \text{gr}_{u\tau+a}$ for every $\tau \geq 0$; equivalently the matching Berlekamp operator $\int_u^a$ descends. For integer $u = n$, the shifted thermal height $h = \tau + 1$ simply scales as $h \mapsto nh$. Thus the mirror is not purely notational: the game filtration has a genuine family of numeric regrading transports.

The final pass proves that the remaining ring hope cannot be repaired by adding residues. First, the temperature-zero residue contains the nonzero class $s = [\ast]$ with $s + s = 0$. Any full dyadic coefficient object has an invertible element π_2 representing 2—either ordinary 2 or its homogeneous initial form—and lift-compatible action sends s to the initial class of $\ast + \ast = 0$; the inverse of π_2 hence forces $s = 0$. This rules out an arbitrary internal product, not merely the Norton candidate. Second, the numeric Norton degree maps have the exact composition defect

$$\Delta(u, v) = v(1 - \delta_u) - \delta_v + \delta_{uv} \geq 0,$$

which is often positive; for $u = 1/2$ and $v = 2$ it equals 1. Hence those external transports cannot be a dyadic scalar action either, and an enrichment cannot repair the mismatch because it already

occurs in exact temperature degrees. The Newton-style one-object reading is impossible under the natural coefficient, associativity, and faithfulness axioms.

Claim levels. **[PROVED]** – proof from standard short-game identities plus the live implementation’s definitions; **[TESTED]** – verified by the commands in Appendix A; **[STANDARD MATH]** – standard thermography/Newton-polygon math used by the repo. The main dichotomy is resolved; the explicitly listed escape hatches are boundaries of the theorem, not conjectural counterexamples.

1 The two tropical objects currently in the code

[STANDARD MATH] The place axis is the usual non-Archimedean tropicalization. In `src/scalar/valued.rs`, a valued scalar satisfies

$$v(xy) = v(x) + v(y), \quad v(x + y) \geq \min(v(x), v(y)),$$

with equality when $v(x) \neq v(y)$. The adaptor `tropicalize` lands in the min-plus semiring. In `src/scalar/newton.rs`, $\text{NP}(f)$ is the lower convex hull of the points $(i, v(a_i))$, and its side slopes are the negatives of root valuations. The Springer tests in `src/forms/springer/local.rs` check that the Newton polygon forgets only the residue square-class data: its slopes are exactly the valuation layers.

[STANDARD MATH] The game axis is also tropical, but internally. A short game’s thermograph consists of a left wall $L_G(t)$, a right wall $R_G(t)$, the mast $\text{mean}(G)$, and the temperature $\text{temp}(G)$; the implementation lives in `src/games/thermography.rs`. The tropical naming layer makes the left wall the max-plus fold over Left options’ right walls, the right wall the min-plus fold over Right options’ left walls, and cooling ordinary addition of the temperature parameter, i.e. tropical multiplication. The test `via_tropical_matches_thermograph` pins this naming equal to the ordinary thermograph.

Remark 1. This is already enough to justify the slogan “thermography is tropical arithmetic”. It is not enough to make thermographs compose under disjunctive sum. Newton polygons have Dumas additivity under polynomial multiplication. Thermographs do not have an analogous strict additivity under game sum.

2 The thermograph is not a sum invariant

Definition 1. For a short game G , write

$$\text{Th}(G) = (L_G, R_G, \text{mean}(G), \text{temp}(G))$$

for the exact thermograph computed by `thermograph`. A thermograph-level sum law would be any operation Φ such that

$$\text{Th}(G + H) = \Phi(\text{Th}(G), \text{Th}(H))$$

for all games in the domain of ordinary thermography.

Proposition 1 (No thermograph-level composition law). **[PROVED]** *The map $G \mapsto \text{Th}(G)$ is not a congruence for disjunctive sum. Consequently, no operation whose inputs are only $\text{Th}(G)$ and $\text{Th}(H)$ can determine $\text{Th}(G + H)$ for all short games.*

Proof. Use the standard games $*$ = $\{0 \mid 0\}$ and $\uparrow$ = $\{0 \mid *\}$. The live thermography engine, and standard temperature theory, give

$$\text{Th}(*) = \text{Th}(\uparrow) = (0, 0, \text{mean} = 0, \text{temp} = 0),$$

where both walls are the constant zero wall.

Now add the same partner $*$. Since $*$ is the number of order two,

$$* + * = 0,$$

so the result is a number and the repo reports temperature -1 for the cold numeric case. But $\uparrow + *$ is still all-small and non-numeric; its thermograph has temperature 0 and constant zero walls. Thus the two input pairs

$$(\text{Th}(*), \text{Th}(*)) \quad \text{and} \quad (\text{Th}(\uparrow), \text{Th}(*))$$

are identical, while the output thermographs differ. This contradicts the existence of Φ . $\square$

Corollary 1. [PROVED] *The pair $(\text{mean}(G), \text{temp}(G))$ cannot be the game-side analogue of $(\text{ac}(x), v(x))$ for a valued field element. It forgets the leading thermic residue that decides cancellation.*

Remark 2. The witness is deliberately low-tech. It lives at temperature zero, before any subtle sidling or nested hot-game pathology appears. The obstruction is not an implementation artifact of complicated thermographs; it is present in the first infinitesimal layer.

3 What survives: temperature as a tropical valuation

[STANDARD MATH] Temperature does satisfy the valuation-shaped inequality

$$\text{temp}(G + H) \leq \max(\text{temp}(G), \text{temp}(H)),$$

with numbers treated as colder than all infinitesimal/hot games. Reversing the order by $v_T(G) = -\text{temp}(G)$ gives the familiar valuation form

$$v_T(G + H) \geq \min(v_T(G), v_T(H)).$$

The equal-temperature case is the game-side vanishing locus. It can cancel downward, or it can retain the same leading temperature.

[TESTED] In this checkout I probed a small catalogue consisting of integers, $*$, $\uparrow$, $\downarrow$, $*2$, switches $\{a \mid b\}$ for small integer a, b , and one-option nested hot games. Across 324 pairwise sums there were no violations of $\text{temp}(G + H) \leq \max(\text{temp } G, \text{temp } H)$. Every unequal-temperature pair hit equality. Equal-temperature pairs produced the expected downward spread:

$$\begin{aligned} \text{temp}(*) &= \text{temp}(*) = 0, & \text{temp}(* + *) &= -1, \\ \text{temp}(*) &= \text{temp}(\uparrow) = 0, & \text{temp}(* + \uparrow) &= 0, \\ \text{temp}(\{1 \mid -1\}) &= \text{temp}(\{2 \mid 0\}) = 1, & \text{temp}(\{1 \mid -1\} + \{2 \mid 0\}) &= -1. \end{aligned}$$

Other equal-temperature probes at input temperature 2 produced output temperatures $-1, 0, 1$, and 2.

Remark 3. This is exactly the tropical-hyperfield shape if one keeps only the scalar temperature: unequal leading values add strictly by max-plus; equal leading values may cancel to any lower layer. The problem is that this scalar law is too coarse to recover walls, masts below the leading layer, or the resulting game value.

4 Switches versus one-sided Newton polygons

[TESTED] There is a clean but shallow dictionary for switches. For $m \in \mathbb{Q}$ and $\tau > 0$, the switch

$$S_{m,\tau} = \{m + \tau \mid m - \tau\}$$

has

$$L_S(t) = m + \tau - t, \quad R_S(t) = m - \tau + t \quad (0 \leq t \leq \tau),$$

and both walls freeze to m at $t = \tau$.

The Newton polygon of a binomial $x^n - \pi^k$ has one side from $(0, k)$ to $(n, 0)$, hence one slope and one root-valuation layer. After an affine change of axes, one wall of $S_{m,\tau}$ is exactly such a one-side polygon; the other wall is its dual reflection. This realizes the requested “one-object probe” only in the trivial one-parameter family.

Proposition 2. [PROVED] *The switch/binomial dictionary does not extend to a thermograph-level sum theorem.*

Proof. The dictionary sends one switch to one linear Newton side. Polynomial multiplication concatenates Newton slope data by Dumas additivity. But Proposition 1 shows that disjunctive sum cannot be read from thermographs alone, even before leaving the zero-temperature layer. Any extension would need extra residue data not present in the switch wall. $\square$

5 The residual bridge: an associated graded game group

[PROVED] The positive residual object begins with the temperature filtration of the game group. Let $F_{<0}$ be the subgroup of short numeric games. For $\tau \geq 0$, set

$$F_{\leq\tau} = F_{<0} \cup \{G \text{ nonnumeric} : \text{temp}(G) \leq \tau\}, \quad F_{<\tau} = F_{<0} \cup \bigcup_{0 \leq \sigma < \tau} F_{\leq\sigma},$$

with the union empty when $\tau = 0$. The standard temperature inequality says these are additive subgroups. The associated graded group is

$$\text{gr}_T(\text{Games}) = \bigoplus_{\tau} F_{\leq\tau} / F_{<\tau}.$$

The leading thermic residue of G would be the class of G in the quotient at $\tau = \text{temp}(G)$, not the pair $(\text{mean}(G), \text{temp}(G))$.

[TESTED] At $\tau = 0$, this is already visible. The thermograph collapses $*$, $\uparrow$, $\downarrow$, and $*2$ to the same constant-zero object. Atomic weight, implemented in `src/games/atomic_weight.rs`, recovers one additive residue on all-small games:

$$aw(\uparrow) = 1, \quad aw(\downarrow) = -1, \quad aw(*) = 0.$$

But atomic weight still has a kernel containing nimber-like residues such as $*$, and $* + * = 0$ shows that this kernel matters. So even the first graded piece is not “the mast”. It is a genuine residual game object.

Remark 4. This is the closest analogue to the valued-field picture. The Newton polygon records valuations and forgets residues; the initial forms in $\text{gr}_v(K[x])$ restore them. The thermograph records temperatures and walls but forgets thermic residues; a graded game object would restore them. The difference is that $\text{gr}_v(K)$ is a graded ring, while short games under disjunctive sum are only an abelian group.

6 The Norton/overheating descent test

[**TESTED**] After the first pass, the repo gained `src/games/heating.rs`: game-valued heating, Norton multiplication $G.U$ by a positive unit U , and Berlekamp overheating $\int_s^t G$. This removes the old infrastructure excuse. We can ask directly whether these operators are well-defined on the associated-graded quotient.

For a fixed layer $\tau \geq 0$, write

$$G \equiv_\tau H \iff \text{temp}(G - H) < \tau.$$

At $\tau = 0$, the lower subgroup $F_{<0}$ is the cold numeric subgroup. Thus $*$ and $* + 1$ have the same residue in $F_{\leq 0}/F_{<0}$, since

$$* - (* + 1) = -1, \quad \text{temp}(-1) = -1 < 0.$$

Proposition 3 (Unrestricted Norton descent fails). [**TESTED**] *There exists a positive nonnumeric unit whose Norton multiplication does not descend to the naive temperature associated graded.*

Proof. Let $U = \uparrow$, which is strictly positive and has temperature 0. The live implementation reports

$$\text{temp}(*.U) = 0, \quad \text{temp}((* + 1).U) = 0,$$

but

$$\text{temp}((*U) - ((* + 1).U)) = 0, \quad \text{aw}((*U) - ((* + 1).U)) = -1.$$

The output difference is therefore not in the lower subgroup $F_{<0}$; it is a leading temperature-zero residue. So two equivalent representatives in the $\tau = 0$ quotient produce inequivalent outputs.

The mechanism is exactly the danger one would expect from the definition: Norton multiplication uses U on integer leaves. The hidden cold integer $+1$, invisible before passing to the quotient, becomes the visible temperature-zero game $\uparrow$ after applying the operator. $\square$

[**TESTED**] The same obstruction appears for overheating. With lower unit $s = \uparrow$ and upper shift $t = 0$,

$$\text{temp}\left(\int_{\uparrow}^0 * - \int_{\uparrow}^0 (* + 1)\right) = 0, \quad \text{aw}\left(\int_{\uparrow}^0 * - \int_{\uparrow}^0 (* + 1)\right) = -2.$$

Thus $\int_s^t$ also fails to descend when the lower unit is allowed to be a nonnumeric positive game.

[**TESTED**] The bounded probe `experiments/under_descent.py` records this explicit witness and also runs the positive numeric-unit audit used below. On a 35-game catalogue and six units it checks 210 exact thermic predictions plus 24 representative pairs each for Norton multiplication and matching overheating. It finds no numeric-unit failure. This points to the exact boundary proved next: nonnumeric units can see cold coefficients erased by the naive quotient, while every positive numeric unit descends individually.

7 Positive result: numeric Norton units descend

The bounded scan above left one conspicuous possibility alive. It is not an accident of the small catalogue; numeric units admit an exact formula.

Definition 2. Let $U = u > 0$ be a short-game number, written in lowest dyadic form $u = m/2^k$. Put

$$\delta_U = \begin{cases} 2^{-k}, & k > 0, \\ 1, & k = 0, \end{cases} \quad a_U = u - \delta_U.$$

Thus the canonical options of a nonintegral u are $u - \delta_U$ and $u + \delta_U$, while a positive integer has the sole Left option $u - 1$.

Proposition 4 (exact numeric Norton thermics). **[PROVED]** *Let $U = u > 0$ be dyadic and write $a = a_U$. For every short game G in the finite thermographic domain,*

$$\text{mean}(G.u) = u \text{mean}(G).$$

If G is not a number, then

$$\text{temp}(G.u) = u \text{temp}(G) + a. \quad (1)$$

If $G = x$ is a number and $\epsilon_x = 1/\text{den}(x)$ is its canonical dyadic mesh, then

$$\text{temp}(x.u) = \begin{cases} -1, & a - u\epsilon_x < 0, \\ a - u\epsilon_x, & a - u\epsilon_x \geq 0. \end{cases} \quad (2)$$

In particular, the image of every cold number has temperature strictly less than a .

Proof. For the canonical numeric unit u , every Norton increment $U + I$ is the same number a : a Left option contributes $U^L = u - \delta_U = a$, and a Right option contributes $2U - U^R = u - \delta_U = a$. Hence the defining recursion is

$$n.u = nu \quad (n \in \mathbb{Z}), \quad G.u = \{G^L.u + a \mid G^R.u - a\} \quad (G \notin \mathbb{Z}). \quad (3)$$

We first need the numeric-option lemma hidden by the integer base case in (3). If x has exact denominator 2^j , induction on j gives mast ux . Integers are literal leaves and have temperature -1 . If $j > 0$, the canonical options are $x \pm 2^{-j}$ and have smaller denominator. Once their images have frozen, the parent walls meet at $s_x = a - u2^{-j}$. When $a = 0$ (precisely $u = 1$ or $u = 2^{-k}$), the simplicity theorem instead makes every $x.u$ the number ux ; when $a > 0$, one has $s_x \geq 0$ for every $j > 0$. Thus (2) follows, together with the strict bound $\text{temp}(x.u) < a$. In particular, Norton multiplication treats integer leaves specially, not all number leaves: for example $(1/2).2 = 1 + *$ has temperature 0.

Now let G be nonnumeric and induct on its birthday. At a scaffold parameter $s = a + ut$, $t \geq 0$, (3), translation of walls, and the induction hypothesis give

$$L_{G.u}(a + ut) = uL_G(t), \quad R_{G.u}(a + ut) = uR_G(t). \quad (4)$$

Numeric options cause no exception: by (2) they have already frozen strictly below a , with mast equal to their ordinary product by u . Equation (4) locates a meeting at $a + u \text{temp}(G)$; it remains to rule out a premature meeting below a .

Write $D_G(t) = L_G^{\text{raw}}(t) - R_G^{\text{raw}}(t) - 2t$. The wall-slope bounds make D_G nonincreasing. If $\text{temp}(G) > 0$, then $D_{G.u}(a) = uD_G(0) > 0$, so monotonicity rules out every earlier zero. Only the infinitesimal case $\text{temp}(G) = 0$ needs care. We use the following elementary stop lemma.

No-premature-meeting lemma. Let $a > 0$, let m be a number, and let X be canonical. If the transformed right wall of $X.u$ has maximal slope $+1$ up to height a and terminates at um , then

Right wins $X - m$ moving first. The mirror statement holds for a transformed left wall of slope -1 .

For completeness, prove the lemma by induction on the birthday of X . The maximal slope forces an extremizing Right option Y . If Y is numeric, the numeric-option lemma and its strict freeze below a force $Y = m$, giving the winning move immediately. Otherwise every Left reply Z of Y has $\text{RS}(Z) \leq m$. Strict inequality is already a Right win; equality forces the same maximal-slope condition and invokes the induction hypothesis. Number avoidance ensures that these are all relevant replies in $Y - m$. This proves the lemma; negation proves its mirror.

Suppose now that both transformed walls met on a terminal interval below a . Put $m = \text{LS}(G) = \text{RS}(G)$. Every stop-maximizing Left option then satisfies the first half of the lemma, while lower-stop options are immediate Right wins. Number avoidance therefore gives $G \leq m$; the mirror argument gives $G \geq m$. Hence $G = m$, contrary to G being nonnumeric. There is no premature meeting. Equation (4) therefore gives exactly $\text{temp}(G.u) = a + u \text{temp}(G)$ and common mast $u \text{mean}(G)$. $\square$

Corollary 2 (numeric Norton descent). **[PROVED]** *For every $\tau \geq 0$, Norton multiplication by a fixed positive dyadic unit u induces an additive map*

$$\overline{A}_{u,\tau} : F_{\leq \tau} / F_{< \tau} \longrightarrow F_{\leq u\tau + a_U} / F_{< u\tau + a_U}. \quad (5)$$

The same map is the generalized overheating operator $\int_u^{a_U}$. For a positive integer n , the shifted degree $h = \tau + 1$ satisfies

$$h(G.n) = n h(G).$$

Proof. Norton multiplication by a fixed positive unit is additive, a standard Norton product theorem [1]. If $G - H$ is nonnumeric of temperature $\sigma < \tau$, (1) gives

$$\text{temp}((G - H).u) = u\sigma + a_U < u\tau + a_U.$$

If $G - H$ is a number, (2) instead puts its image strictly below a_U , hence strictly below the target layer. Thus the map is independent of the chosen representative and is additive. Finally, (3) is exactly Berlekamp overheating from lower unit u to upper shift a_U . For $u = n$, $a_U = n - 1$, so $\text{temp}(G.n) + 1 = n(\text{temp}(G) + 1)$ on every nonnumeric graded layer. $\square$

[TESTED] Proposition 4 is computed without materializing the Norton product by the two Rust helpers

`numeric_norton_regrade` `numeric_norton_mean_temperature`.

Source-level tests compare them with the actual recursive product on integer and nonintegral numeric games, all-small games, switches, and a nested hot game for eight units from $1/4$ through 3. The Python probe independently checked 210 game/unit thermic pairs and 24 representative pairs for both Norton multiplication and the matching overheating operator, with no numeric-unit failure.

Remark 5. This is the first nontrivial positive transport in the **under** thread. It is weaker than a ring structure: u is an external numeric coefficient, not an arbitrary game, and (5) regrades rather than multiplying two residual game classes. It is not even a multiplicative action of the positive dyadics: $A_2 A_{1/2}(\ast)$ has temperature 1, whereas $A_1(\ast) = \ast$ has temperature 0. The result is a family of additive transports. That is still stronger than the scalar temperature hyperoperation: each transport acts on the residual classes themselves and, for integer units, multiplies thermal height exactly as a ramification index multiplies a valuation scale.

8 The exact Norton composition defect

Write $A_u(G) = G.u$ for Norton multiplication by the positive dyadic unit u , and let

$$r_u(\tau) = u\tau + a_u, \quad a_u = u - \delta_u, \quad \delta_u = \frac{1}{\text{den}(u)}.$$

Corollary 2 says that A_u induces an additive map with degree function r_u . Associativity of an ordinary dyadic scalar action would require $A_v A_u = A_{uv}$ and hence $r_v r_u = r_{uv}$. The failure is exact.

Proposition 5 (composition defect). **[PROVED]** *For positive dyadics u, v and every $\tau \geq 0$,*

$$r_v(r_u(\tau)) = r_{uv}(\tau) + \Delta(u, v), \quad \Delta(u, v) = v(1 - \delta_u) - \delta_v + \delta_{uv} \geq 0. \quad (6)$$

The zero-defect pairs are exactly:

- (i) u nonintegral and $v = 2^{-\ell}$ for some $\ell \geq 0$;
- (ii) u an odd positive integer and arbitrary v ;
- (iii) u an even positive integer and v an integer.

Proof. Composition of the affine maps gives

$$r_v(r_u(\tau)) - r_{uv}(\tau) = v(u\tau + u - \delta_u) + v - \delta_v - (uv\tau + uv - \delta_{uv}),$$

which is (6).

If u is an integer, then $\delta_u = 1$ and $\Delta = \delta_{uv} - \delta_v$. Multiplication by an integer cannot increase a dyadic denominator. Equality holds for integral v , and for nonintegral v exactly when u is odd; an even u cancels at least one denominator factor.

Now let $u = m/2^k$ in lowest terms with $k > 0$. If v is nonintegral, both numerators are odd, so $\delta_{uv} = \delta_u \delta_v$ and

$$\Delta = (1 - \delta_u)(v - \delta_v).$$

This is nonnegative and vanishes exactly when v has numerator 1, i.e. $v = 2^{-\ell}$. If $v = n$ is an integer, write $n = 2^s q$ with q odd. For $s \geq k$, $\delta_{uv} = 1$ and $\Delta = n(1 - 2^{-k}) > 0$. For $s < k$,

$$\Delta = n - 1 - \frac{n - 2^s}{2^k};$$

this is zero at $n = 1$ and positive for $n > 1$ (separate $s = 0$, or bound $(n - 2^s)/2^k \leq (n - 2^s)/2$ when $s > 0$). This proves both nonnegativity and the classification. $\square$

Corollary 3 (Norton is not a dyadic action). **[PROVED]** *No temperature-preserving residue refinement can make all numeric Norton transports into the scalar action of an associative algebra whose dyadic coefficients multiply ordinarily.*

Proof. Take $u = 1/2$, $v = 2$, and $G = *$. Proposition 5 gives $\Delta = 1$, and the games themselves give

$$\text{temp}(A_2 A_{1/2}(*)) = 1, \quad \text{temp}(A_1(*)) = \text{temp}(*) = 0.$$

Associativity would identify these two outputs. A residue refinement cannot do so while preserving the temperature grading: the two leading classes occur in different exact degrees. $\square$

The helper `numeric_norton_composition_defect` implements (6) without building either product. The bounded Python audit checks 2,304 positive-dyadic pairs for nonnegativity and the exact zero classification, then pins five selected values against recursively materialized Norton products.

9 The coefficient-torsion obstruction

The preceding theorem closes the standard Norton route. A stronger argument rules out replacing Norton multiplication by an arbitrary internal product. We first state the naturality contract, because without one an unrelated binary operation could always be invented.

Definition 3 (full-dyadic Newton-style coefficient action). A full-dyadic Newton-style coefficient action on the temperature filtration is a residue-enriched additive object R and a unital graded coefficient ring C such that:

- (i) its addition is induced by disjunctive sum;
- (ii) the thermic residues embed faithfully, in particular the nonzero class $[*] \in F_{\leq 0}/F_{< 0}$ remains nonzero;
- (iii) C acts unital-associatively and distributively on R ;
- (iv) C has mutually inverse homogeneous elements $\pi_2, \pi_{1/2}$ representing 2 and $1/2$; and
- (v) filtered-scalar compatibility holds: if $d_2 = \deg(\pi_2)$, then

$$\pi_2[G]_\tau = [G + G]_{\tau+d_2}.$$

Here a common abelian grading group Γ receives fixed embeddings of the thermic and C -gradings, $\tau + d_2$ is their sum in Γ , and the right side denotes the initial class in that shifted component (hence is zero when $G + G$ drops strictly below it).

This is the explicit coefficient-compatibility contract tested here. It includes an ordinary unital $\mathbb{Z}[1/2]$ -algebra by taking $\pi_2 = 2$, and it also includes the graded coefficient field of a valued dyadic field by taking $\pi_2 = \text{in}(2)$ and $\pi_{1/2} = \text{in}(1/2)$. It does *not* claim that the initial-form map from a valued field to its associated graded is additive. Requiring the scalar maps also to be the Norton maps is stronger, and was already ruled out by Corollary 3.

Proposition 6 (dyadic torsion no-go). **[PROVED]** *No faithful full-dyadic Newton-style coefficient action exists. In particular, no internal game algebra carrying such coefficients exists.*

Proof. In the live filtration convention, games of negative temperature are precisely the cold numbers. The game $*$ is nonnumeric with temperature 0, so

$$s = [*] \neq 0 \quad \text{in} \quad F_{\leq 0}/F_{< 0}.$$

But $* + * = 0$, hence filtered-scalar compatibility gives $\pi_2 s = [0]_{d_2} = 0$ in the shifted target degree. Since π_2 has inverse $\pi_{1/2}$, associativity gives

$$s = (\pi_{1/2}\pi_2)s = \pi_{1/2}(\pi_2 s) = 0,$$

contradicting faithful retention of the residue. No commutativity assumption is needed. The same argument kills every nonzero finite-number residue, since each is self-inverse under disjunctive sum. $\square$

Remark 6. The raw gr_T does not itself contain the dyadic numbers: they lie in the cold subgroup below degree zero. Proposition 6 blocks both the repair “retain cold coefficients as a unital $\mathbb{Z}[1/2]$ subring” and a secondary valuation/Rees repair in which the *full dyadic field* acts by initial forms: in the latter, $\text{in}(2)$ remains a homogeneous unit with inverse $\text{in}(1/2)$. One can evade the theorem by

retaining only a valuation-ring or integer action in which 2 is not inverted, by discarding the dyadics, by killing the nimber residue, or by abandoning a shared unit, associativity, or distributivity. A characteristic-2 ring on a nimber-only slice is therefore possible in principle. None of these is a single full-dyadic Newton-style ring for the full thermographic filtration: each drops one of the two coefficient/residue worlds that the proposed bridge was meant to unify.

10 Resolution: two objects, a substantive shadow

[PROVED] The original dichotomy is now decided under the explicit naturality contract above. The place axis has a valued ring and hence an associative graded ring; Dumas additivity and Newton slopes use that multiplication. The game axis has a temperature-filtered abelian group. Its degree-zero residue has essential 2-torsion, so it admits no coefficient-compatible action by the full dyadic field, even through graded initial forms; its standard product-like numeric operators have the independent composition defect (6). Thermography and Newton polygons are therefore *two tropical objects*, not one tropical ring with the full cold coefficient world.

The connection is nevertheless mathematical rather than typographical:

- temperature obeys the non-Archimedean max inequality and its scalar hyperaddition records the equal-degree cancellation locus;
- the exact thermograph recursion uses dual max-plus/min-plus wall folds;
- switches and binomial Newton sides share the bounded one-parameter dictionary; and
- every positive numeric Norton unit induces the exact external affine regrading (5).

What fails is precisely the multiplicative unification that would turn those shadows into one associated graded ring. Describing individual residue groups, or studying characteristic-2 and nonassociative structures on restricted slices, remains worthwhile follow-up mathematics; it is no longer needed to answer the **under** problem posed here.

A Checks run in this pass

The code-level facts used above were checked against the live repo with:

- `cargo test tropical -- --nocapture`: 15 relevant tests passed, including tropical semiring laws and `thermograph_via_tropical`.
- `cargo test newton -- --nocapture`: 8 Newton-polygon tests passed, including Eisenstein slopes and Dumas additivity.
- `cargo test springer -- --nocapture`: 50 Springer/local-global tests passed, including the Newton-polygon/Springer-shadow check.
- A Python probe through the installed `.venv` binding checked 324 small-game sums for the temperature inequality and the off-diagonal equality pattern, and printed the witnesses in Proposition 1.
- `cargo test heating`: 15 heating/Norton/overheating tests passed, including the explicit non-descent witness `hot_units_do_not_descend_mod_cold_numbers`, the exact dyadic regrade, the load-bearing numeric-image lemma, matching-overheating equality, numeric quotient descent, a bounded day-three singleton-option census, a Norton-linearity sentinel, and the exact composition-defect theorem.

- `cargo check --features python`: the new Python thermic shortcuts compile against the feature-gated bindings.
- `python experiments/under_descent.py`: recorded the $*$ versus $* + 1$ obstruction for unit $\uparrow$; checked 210 exact thermic predictions on a 35-game catalogue and six numeric units; and checked 24 representative pairs each for Norton multiplication and matching overheating, with zero numeric-unit failures. It also checked 2,304 positive-dyadic pairs for nonnegative defect and the exact zero classification, with five game-level composition witnesses.

References

- [1] Berlekamp, Conway, and Guy. *Winning Ways for your Mathematical Plays*.
- [2] Aaron Siegel. *Combinatorial Game Theory*.
- [3] Elwyn Berlekamp. The economist’s view of combinatorial games. In *Games of No Chance*, 1996.
- [4] Elwyn Berlekamp. Blockbusting and Domineering. *Journal of Combinatorial Theory, Series A* 49 (1988), 67–116.
- [5] Oleg Viro. Hyperfields for tropical geometry I.
- [6] Diane Maclagan and Bernd Sturmfels. *Introduction to Tropical Geometry*.